



Ermir Mulaku

FULL-STACK SOFTWARE ENGINEER
TYPESCRIPT · REACT NATIVE · APPLIED AI

CONTACT

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LinkedIn

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Portfolio

www.ermir-mulaku.com

GitHub

github.com/ErmirMulaku

Location

Prishtina, Kosovo

CORE SKILLS

TypeScript & Web

TypeScript, React 19, Next.js 15/16, Redux Toolkit, TanStack Query, SCSS/Tailwind, i18n & RTL, PWA, Core Web Vitals

Mobile & Cross-Platform

React Native / Expo (OTA, push, native printing), Electron, Socket.IO real-time

APIs & Data

NestJS, Node.js, Laravel/PHP, REST · GraphQL · gRPC, Prisma/PostgreSQL, MySQL, queued jobs, JWT/RBAC, multi-tenant design

Applied AI

OpenAI function calling & embeddings, RAG chatbots, FAISS vector search, LightGBM ranking, Python; Claude Code & Copilot daily

Quality & Production

Jest, Playwright, Cypress, Testing Library, Storybook, GitHub Actions CI/CD, Prometheus, health probes, A/B testing & analytics

Cloud & Ops

AWS (App Runner, ECR, RDS), Google Cloud Run, Vercel, Docker, Kubernetes, Helm

CERTIFICATIONS

Advanced React Component Patterns

Udemy · 2022

Functional Programming in JavaScript

Udemy · 2023

Advanced JavaScript & React

Beetroot Academy · 2021

EDUCATION

BSc, Computer Science

University of Prishtina · 2017–2021

PROFILE

Full-stack product engineer with **5+ years shipping TypeScript products end to end** — Next.js and React Native front ends, NestJS and Laravel APIs, and the PostgreSQL, queue and real-time layers behind them. I build **customer-facing AI in production**: an LLM recommendation chatbot and FAISS vector search on a Swiss education marketplace, an **agentic booking assistant** (OpenAI function calling) whose tool calls are executed and validated server-side, and a learning-to-rank recommender reaching **NDCG@7 0.68 / 79.5% hit-rate@7** over ~50k interactions. I own what I ship after it ships — Jest and Playwright gates, health probes, Prometheus metrics, Core Web Vitals, A/B tests to decide what changes next. I use **Claude Code daily** and stay accountable for every line it helps me write. Based in Prishtina: remote-first, and on hand for in-person days with the Kosovo engineering team.

EXPERIENCE

Software Engineer

Jul 2025 – Present

eduwo · Prishtina, Kosovo (Hybrid)

- Ship production **React 19 / Next.js 15** features in TypeScript on a Swiss education marketplace serving **20,000+ users** — a multi-step matching wizard, course/study/profile journeys and an AI recommendation chatbot — across four locales with SSR and PWA support.
- Built the **customer-facing AI end to end**: a FAISS vector-search service and the recommendation chatbot (service layer → Next.js API route → React UI), plus OpenAI-powered content generation in the admin.
- Built a **learning-to-rank recommender** in Python (LightGBM, OpenAI embeddings, probability calibration) reaching **NDCG@7 0.68 and 79.5% hit-rate@7** on ~50k user–course interactions.
- Design, build and document **REST APIs in Laravel** — auth, validation, queued background jobs, transactional email and notifications — consumed by the web app, the admin and native clients.
- Own quality and uptime** for the services I ship: ESLint, Jest, Playwright E2E, pre-commit gates and monitoring of my endpoints in production.

Frontend Developer

Jul 2021 – Jul 2025

eduwo · Kosovo

- Built complex single-page applications with React, Redux Toolkit and TypeScript, with a hard bar on performance, responsiveness and UX detail; tracked Core Web Vitals.
- Ran **data-driven UI experiments** — A/B testing plus product analytics — and used production data to decide what to build and fix next.
- Designed and documented **REST API contracts** powering frontend features, coordinating payloads with backend and native consumers; built Next.js API routes as a BFF layer.
- Implemented interfaces from scratch in SCSS — responsive layouts, animation, carousels, charts, maps, PWA — across the modern JS toolchain (Webpack, esbuild/SWC, ESLint, Prettier, Formik + Yup).

Intern Front-End Developer

Feb 2021 – Jun 2021

Kutia · Prishtina, Kosovo

- Built responsive, accessible UI components inside a cross-functional product team, strengthening fundamentals in JavaScript, React, HTML5, CSS3 and Git workflows.

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SELECTED LINKS

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WORKING STYLE

End to end

I shape the problem with product and design, build it across the stack and follow it into production.

Production first

Instrumented, tested and observable — health probes, metrics, E2E gates, and the metrics that say whether it worked.

AI-native

Claude Code and Copilot in the loop every day; the judgement and the review stay mine.

SHIPPED PRODUCTS

TutorHub — Full-Stack Tutoring Marketplace (independent)

[NestJS](#) · [GraphQL](#) · [gRPC](#) · [Prisma/PostgreSQL](#) · [Next.js](#) · [Expo](#) · [Elixir/Phoenix](#) · [AWS](#) · [Storybook](#)

- Designed and built a marketplace monorepo end to end: a **NestJS API exposing one domain over REST, GraphQL and gRPC**, a Next.js SSR/PWA storefront, a React/Redux tutor dashboard and an **Expo mobile app**.
- Shipped an **OpenAI function-calling booking assistant** with server-executed, validated tool calls — the model proposes, the API enforces — plus an Elixir/Phoenix gRPC microservice for lesson-reminder notifications.
- Published an **open-source npm package** (timezone-aware lesson-availability engine) and a shared, RTL-safe UI library documented in Storybook.
- Automated **CI/CD with GitHub Actions** (lint, typecheck, test, build, Lighthouse budget) deploying the API to AWS App Runner via ECR and the front ends to Vercel; exposed **health/readiness probes and Prometheus metrics**.

Live: tutorhub-marketplace.vercel.app · **npm:** [tutorhub-slot-engine](https://www.npmjs.com/package/tutorhub-slot-engine) · **Storybook:** tutorhub-storybook.vercel.app

S2O — Cross-Platform Restaurant Platform · Viviark

[NestJS](#) · [Next.js](#) · [React Native \(Expo\)](#) · [Electron](#) · [Socket.IO](#) · [Kubernetes](#)

- Built a **multi-tenant SaaS end to end**: a NestJS REST API plus three clients — a Next.js web admin, a **React Native (Expo) mobile app** and an Electron desktop POS — sharing one API and a real-time Socket.IO layer.
- Built a **real-time order pipeline** (WebSocket gateways) streaming orders to kitchen and bar displays with **server-enforced status transitions**; dual JWT + 4-digit PIN auth; FIFO inventory with batch and expiry tracking.
- Authored the **multi-tenant migration architecture** — container + database per tenant on Kubernetes, Helm-templated deployments and an automated onboarding control plane — moving the platform from pooled to fully isolated per-restaurant infrastructure.
- Shipped the native clients myself: **Expo OTA updates, push notifications and native printing**, and electron-builder packaging (Windows NSIS, Linux AppImage).

Live: App Store & Google Play — “S2O App” · s2o-menu-theta.vercel.app

Spicy Analytics — Multilingual RTL Web Platform

[Next.js](#) · [i18n](#) · [RTL](#) · [SSR / SEO](#)

- Built a production multilingual platform with **full RTL support** (English LTR + Arabic RTL, mirrored layout) via a key-based i18n system across every page and component, for an enterprise client roster spanning healthcare, banking and telecom.
- Implemented per-locale routing, SSR-rendered SEO metadata and next/image optimization across a responsive, content-heavy site.

Live: spicyanalytics.com

Three60 Music — Animated E-commerce Site

[Next.js](#) · [CSS animations](#) · [E-commerce](#)

- Built an animation-rich, fully responsive Next.js storefront with cart, a two-brand information architecture, SEO/social metadata and CDN-served media.

Live: three60-music.com